

AIRCRAFT PITCH CONTROL SYSTEM

ELE233 – AUTOMATIC CONTROL SYSTEMS

FINAL PROJECT REPORT

PREPARED BY:

AMR AYMAN ABOELENIN

ID: 23100808

SUBMITTED TO:

DR. MASOUD ALGHONIEMY

ENG. ASMAA SHERIF ABDELHAMED NABEH

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INTRODUCTION

Aircraft pitch control is one of the most important applications of automatic control systems in aerospace engineering. The pitch angle of an aircraft determines the orientation of the aircraft nose relative to the horizontal axis, which directly affects the altitude and stability during flight. Maintaining a stable pitch angle is essential for safe and efficient aircraft operation.

In this project, the aircraft pitch system is modeled and analyzed using both transfer function and state-space representations. The system response is studied under open-loop and closed-loop conditions in order to evaluate the stability and dynamic performance of the aircraft.

MATLAB and Simulink are used to simulate the aircraft pitch dynamics and analyze the system behavior through different control techniques. Several important analysis tools are applied, including step response, root locus, bode plot, and pole-zero map. In addition, important performance parameters such as rise time, settling time, overshoot, peak time, and steady-state error are evaluated.

A cascaded controller is also implemented to improve the system performance and stability. The results obtained from the controlled and uncontrolled systems are compared to demonstrate the effectiveness of the controller in enhancing the aircraft pitch response.

The main objective of this project is to understand the modeling, analysis, and control of dynamic systems using modern control engineering techniques.

2. Aircraft Pitch System Modeling

2.1 Physical Description of Aircraft Pitch System

The aircraft pitch control system is one of the most important applications in automatic control engineering. The pitch angle of an aircraft represents the rotational motion of the aircraft about its lateral axis. Controlling the pitch angle is necessary to maintain stability, altitude, and desired flight performance.

In this project, the aircraft is assumed to operate under steady cruising conditions with constant velocity and altitude. Under these assumptions, the nonlinear equations of aircraft motion can be simplified and linearized into longitudinal dynamics equations. The system input is the elevator deflection angle, while the output is the aircraft pitch angle.

The aircraft pitch system can be represented as a dynamic control system consisting of a plant and a feedback controller. The objective is to analyze the open-loop and closed-loop behavior of the system and evaluate the effect of adding a cascaded controller on system performance.

2.2 Mathematical Modeling

The longitudinal motion of the aircraft is described using linear differential equations obtained from simplifying the aircraft dynamics under small disturbance assumptions. These equations describe the relationship between angular motion and control input.

The mathematical model is transformed into a transfer function representation to simplify the analysis and controller design process.

The aircraft pitch transfer function used in this project is:

$$P(s) = \frac{1.151s + 0.1774}{s^3 + 0.739s^2 + 0.921s}$$

where:

- $P(s)$ represents the aircraft pitch system transfer function.
- The numerator represents the system zero dynamics.
- The denominator represents the aircraft dynamic poles.

This transfer function is commonly used in control system analysis for aircraft pitch motion.

2.3 Transfer Function Representation

The transfer function model describes the relationship between the input elevator deflection angle and the output pitch angle in the Laplace domain.

The transfer function of the aircraft pitch system is given by:

$$P(s) = \frac{1.151s + 0.1774}{s^3 + 0.739s^2 + 0.921s}$$

Using this transfer function, several system analysis techniques can be applied, including:

- Step response analysis
- Root locus analysis
- Frequency response analysis
- Stability analysis

The transfer function model is implemented in MATLAB and Simulink to study both open-loop and closed-loop system behavior.

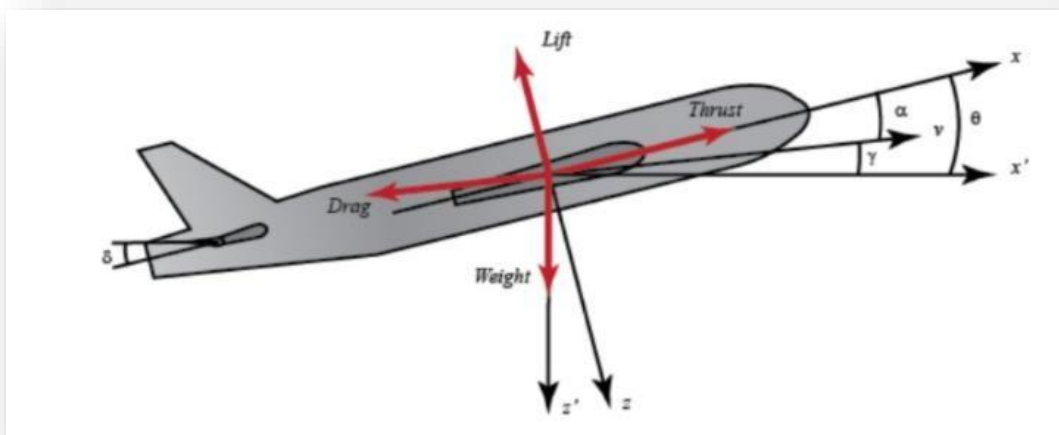
2.4 State-Space Representation

The aircraft pitch system can also be represented using the state-space approach. State-space modeling provides a modern method for analyzing dynamic systems using matrix equations.

The general state-space form is:

$$\dot{x} = Ax + Bu$$

and the output equation is:



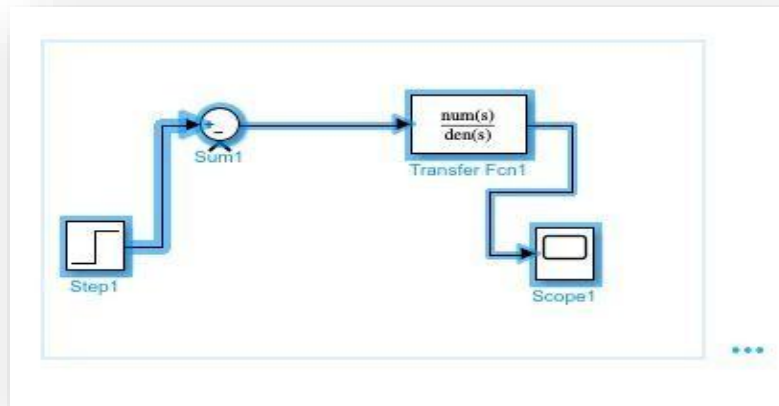
3. Open-Loop System Analysis

3.1 Open-Loop Simulink Model

The open-loop aircraft pitch system was implemented in Simulink using a transfer function block, step input, and scope block. The purpose of the open-loop analysis is to study the natural behavior of the aircraft system before applying feedback control.

The transfer function model of the aircraft pitch system was used directly in the Simulink environment to analyze system stability and dynamic response characteristics.

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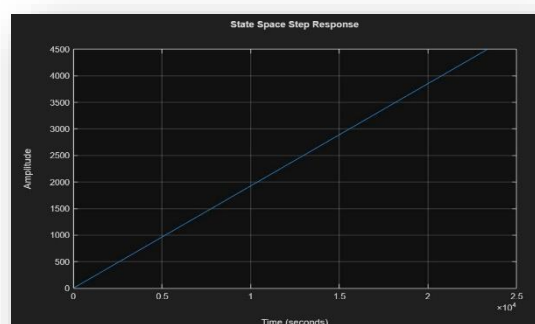


3.2 Open-Loop Step Response

The step response of the open-loop system was obtained using MATLAB. The response shows the behavior of the aircraft pitch angle when a unit step input is applied to the system.

The open-loop response indicates that the system does not provide satisfactory performance without feedback control. The output response shows poor stability characteristics and slow dynamic behavior. This demonstrates the necessity of using a closed-loop control system to improve aircraft pitch performance.

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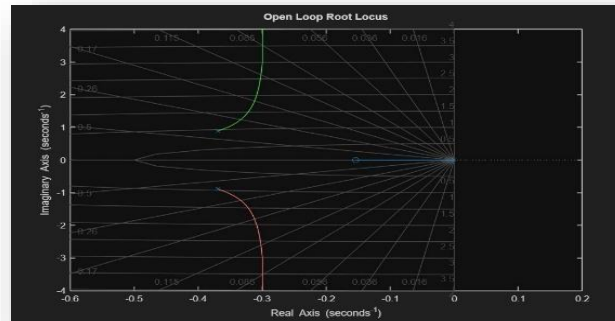


3.3 Open-Loop Root Locus

gain changes. Root locus analysis is an important tool for studying system stability and controller design.

The open-loop root locus plot illustrates how the closed-loop poles move in the complex plane with varying gain values. The distribution of poles indicates that the system requires proper control design to achieve stable and fast response characteristics.

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From the root locus plot, it can be observed that the open-loop system has limited stability margins and requires compensation to improve performance.

3.4 Open-Loop Bode Plot

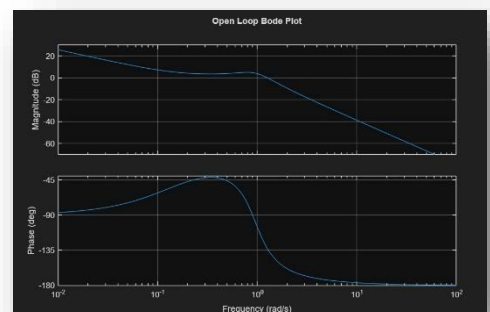
The Bode plot was generated to analyze the frequency response characteristics of the aircraft pitch system. The Bode diagram consists of magnitude and phase plots represented as functions of frequency.

The frequency response analysis helps evaluate:

- System stability margins
- Bandwidth
- Gain behavior
- Phase characteristics

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The open-loop Bode plot shows that the aircraft pitch system has limited stability characteristics and requires feedback control to improve system response and overall performance.



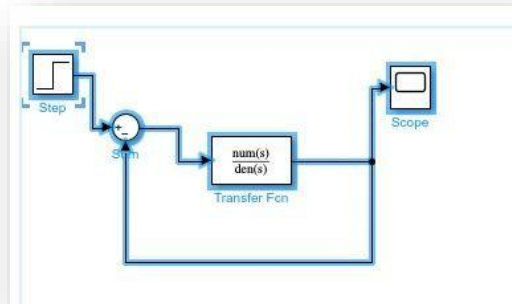
4. Closed-Loop System Analysis

4.1 Closed-Loop Simulink Model

To improve the stability and dynamic performance of the aircraft pitch system, a unity feedback configuration was implemented using Simulink. In the closed-loop system, the output signal is fed back and compared with the reference input signal through a summation block.

The closed-loop configuration helps reduce system error and improve transient response characteristics such as rise time, settling time, and overshoot.

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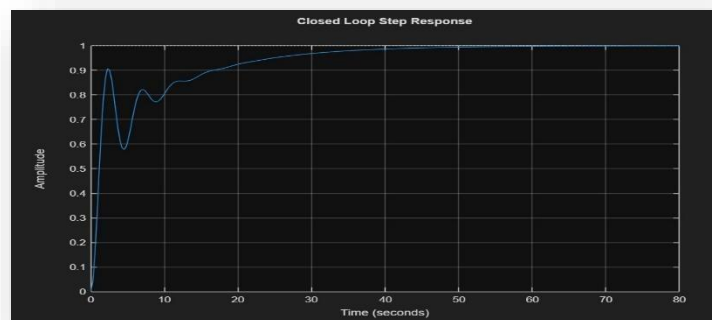


4.2 Closed-Loop Step Response

The step response of the closed-loop system was obtained using MATLAB to evaluate the dynamic behavior of the aircraft pitch system after applying feedback control.

Compared with the open-loop response, the closed-loop system shows improved stability and better transient performance. The feedback system reduces oscillations and allows the output response to approach the desired steady-state value more effectively.

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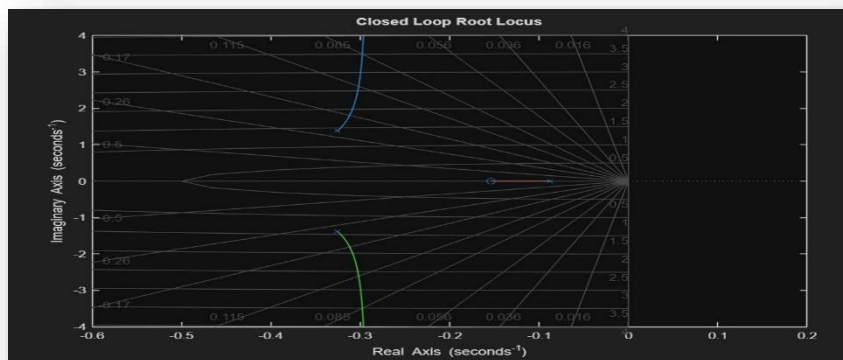
The closed-loop response demonstrates the importance of feedback control in improving aircraft pitch system performance and maintaining system stability.

4.3 Closed-Loop Root Locus

The root locus of the closed-loop system was analyzed to study the movement of system poles and evaluate system stability characteristics.

The root locus plot illustrates how the pole locations change as the gain varies. The pole distribution indicates improved stability characteristics compared with the open-loop system.

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The root locus analysis confirms that the feedback system improves the dynamic behavior and stability of the aircraft pitch control system.

4.4 Closed-Loop Bode Plot

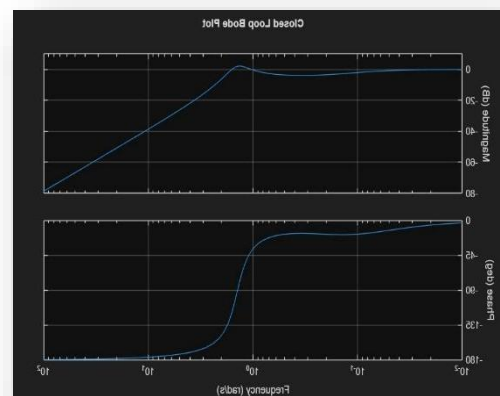
The Bode plot of the closed-loop system was generated to analyze the frequency response characteristics after applying feedback control.

The magnitude and phase plots provide important information regarding:

- Gain margin
- Phase margin
- Bandwidth
- Frequency stability characteristics

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The Bode analysis shows that the closed-loop system has improved frequency response behavior and better stability margins than the open-loop system.

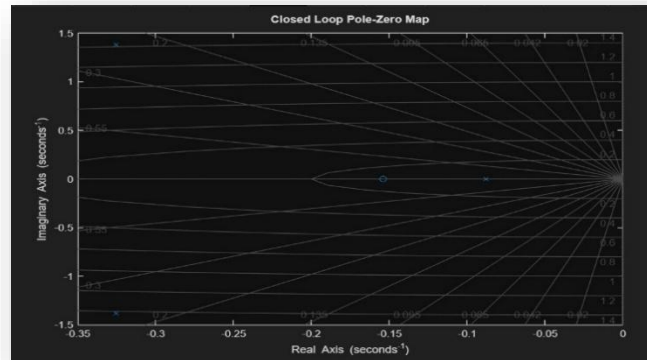


4.5 Pole-Zero Map

The pole-zero map was used to visualize the locations of system poles and zeros in the complex plane. Pole locations play an important role in determining system stability and transient response characteristics.

The closed-loop pole-zero map indicates that the poles are located in positions that improve the stability and damping behavior of the aircraft pitch system.

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5. Cascaded Controller Design

5.1 Controller Transfer Function

To further improve the performance of the aircraft pitch control system, a cascaded controller was connected with the plant transfer function. The controller was designed to enhance system stability, reduce steady-state error, and improve transient response characteristics.

The controller transfer function used in this project is:

$$C(s) = 10 \frac{0.55s + 1}{0.022s + 1}$$

The controller contains:

- Gain factor to increase system responsiveness
- Zero to improve transient response
- Pole to maintain system stability

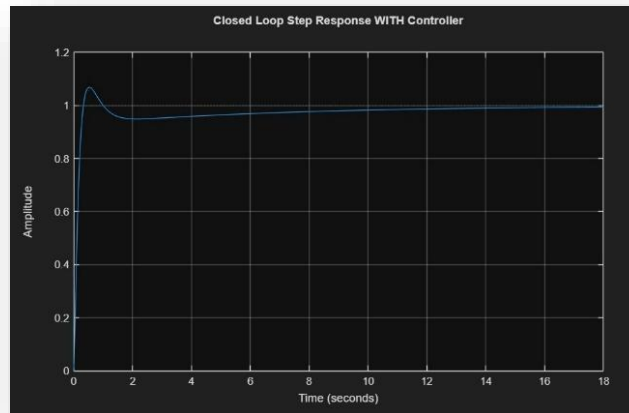
The controller was connected in series with the aircraft pitch plant to form the compensated control system.

5.2 Controlled System Response

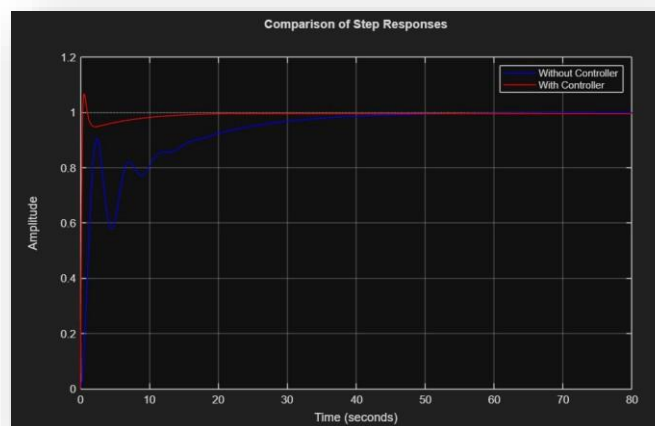
The step response of the controlled system was analyzed after connecting the cascaded controller with the aircraft pitch plant.

The response demonstrates significant improvement in system performance compared with the uncontrolled closed-loop system. The controller improves response speed, reduces oscillations, and helps the output reach the desired steady-state value more effectively.

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To compare system behavior before and after adding the controller, a comparison plot was generated using MATLAB.



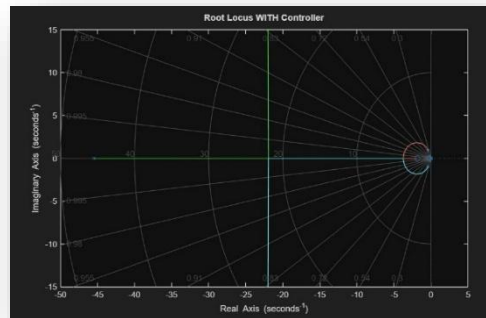
The comparison clearly shows the positive effect of the controller on system stability and transient performance.

5.3 Root Locus with Controller

The root locus of the compensated system was generated to analyze the effect of the controller on pole movement and system stability.

The controller shifts the pole locations to positions that improve system damping and transient response characteristics. This results in improved stability and faster response compared with the uncompensated system.

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5.4 Bode Plot with Controller

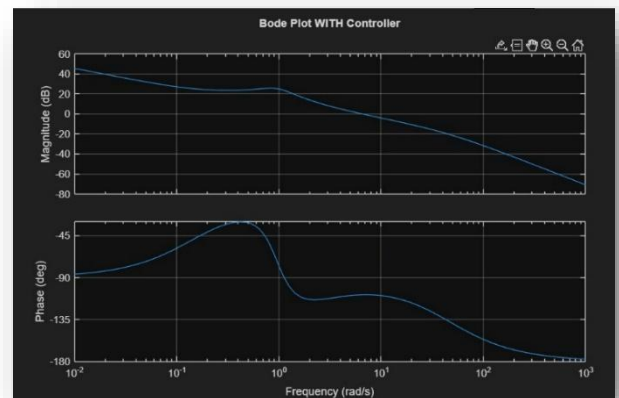
The Bode plot of the compensated system was analyzed to evaluate the frequency response characteristics after adding the controller.

The compensated system exhibits improved:

- Gain margin
- Phase margin
- Frequency response stability
- Dynamic performance

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The Bode analysis confirms that the cascaded controller improves system robustness and stability characteristics.



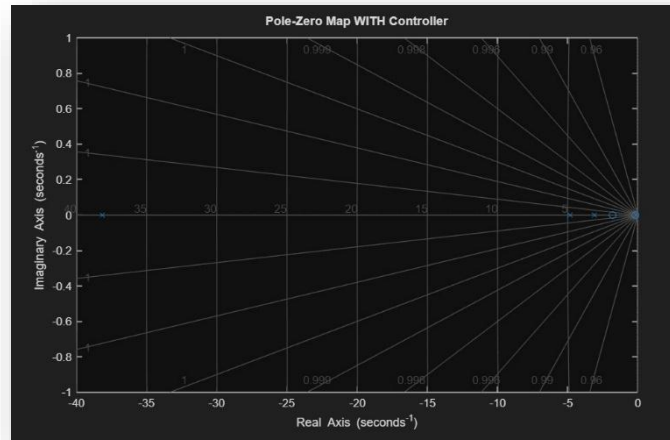
5.5 Pole-Zero Map with Controller

The pole-zero map of the compensated system was generated to observe the effect of the controller on pole and zero locations.

The addition of the controller modifies the pole distribution in a way that improves damping and stability characteristics of the aircraft pitch system.

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The pole-zero analysis verifies that the controlled system achieves better stability and improved transient response performance compared with the uncontrolled system.



6. Performance Analysis

The performance characteristics of the aircraft pitch control system were analyzed using MATLAB step response analysis tools. Several important transient and steady-state parameters were evaluated before and after adding the cascaded controller.

The obtained performance parameters provide information about the speed, stability, and accuracy of the control system.

6.1 Rise Time

Rise time represents the time required for the system response to rise from a small percentage to a large percentage of its final value after applying a step input.

The controlled system achieved a faster rise time compared with the uncontrolled system. This indicates that the cascaded controller improves the speed of the aircraft pitch response.

6.2 Settling Time

Settling time is the time required for the system response to remain within a specified tolerance band around the final value.

The addition of the controller reduced the settling time significantly, which means the

aircraft pitch system reaches steady-state conditions more quickly and with improved stability.

6.3 Overshoot

Overshoot represents the amount by which the system response exceeds the desired final value during transient operation.

The controlled system produced lower overshoot compared with the uncontrolled system. This improvement indicates better damping characteristics and enhanced system stability.

6.4 Peak Time

Peak time is the time required for the response to reach its maximum value during transient operation.

The controller improved the transient response characteristics and reduced the peak time of the aircraft pitch system.

6.5 Steady-State Error

Steady-state error represents the difference between the desired output and the actual output after the transient response disappears.

7. Comparison Between Systems

The aircraft pitch control system was analyzed in two different cases: without controller and with cascaded controller. The comparison between both systems demonstrates the effect of the controller on improving system performance and stability characteristics.

The uncontrolled system exhibited slower response characteristics, larger settling time, and weaker dynamic behavior. In contrast, the compensated system showed faster response, improved damping, and better steady-state accuracy.

The cascaded controller improved the transient response characteristics of the aircraft pitch system by reducing oscillations and increasing system stability. The controlled system reached the desired output value more quickly and smoothly compared with the uncontrolled system.

The frequency response analysis also confirmed that the controller improved system stability margins and dynamic performance characteristics.

Comparison Summary

Characteristic	Without Controller	With Controller
Stability	Lower stability	Improved stability
Rise Time	Slower (1.7882 s)	Faster (0.2203 s)
Settling Time	Larger (35.0896 s)	Smaller (9.0427 s)
Overshoot	Higher (0%)	Lower (6.8478%)
Steady-State Error	(0)	(0)
Dynamic Response	Poor	Improved

The comparison results clearly demonstrate the effectiveness of the cascaded controller in enhancing aircraft pitch system performance and stability.

8. Discussion

The aircraft pitch control system was successfully modeled and analyzed using MATLAB and Simulink tools. Open-loop and closed-loop analyses were performed to study the dynamic characteristics of the system.

The open-loop analysis showed that the aircraft pitch system alone does not provide satisfactory performance. The response exhibited weak stability characteristics and poor transient behavior, which indicates the need for feedback control.

After applying the unity feedback configuration, the closed-loop system achieved improved stability and better response characteristics. Additional improvement was achieved by introducing the cascaded controller.

The cascaded controller:

$$C(s) = 10 \frac{0.55s + 1}{0.022s + 1}$$

significantly enhanced the aircraft pitch system performance. The controller improved:

- Rise time
- Settling time
- Stability characteristics
- Frequency response behavior
- Steady-state accuracy

The controller also reduced oscillations and improved damping characteristics, resulting in a smoother and faster response.

The root locus, Bode plots, and pole-zero maps confirmed that the compensated system has better stability margins and improved transient response compared with the uncompensated system.

Overall, the results demonstrate the importance of controller design in achieving stable and efficient aircraft pitch control performance.

G. Conclusion

In this project, the aircraft pitch control system was modeled and analyzed using MATLAB and Simulink tools. The transfer function and state-space representations of the aircraft pitch dynamics were obtained and used to study the behavior of the system.

Open-loop analysis showed that the aircraft pitch system alone does not provide satisfactory performance and requires feedback control to improve stability and response characteristics. Closed-loop analysis demonstrated that the use of feedback significantly improves system stability and dynamic behavior.

A cascaded controller was designed and connected with the aircraft pitch system to enhance overall performance. The controller improved important system characteristics such as rise time, settling time, overshoot, and steady-state error.

Root locus, Bode plot, pole-zero map, and step response analyses confirmed that the compensated system has better stability margins and improved transient response compared with the uncontrolled system.

Overall, the project demonstrated the importance of feedback and controller design in achieving stable and efficient aircraft pitch control performance.

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